



AQUA

Electric actuators for rotary mixing valves
230 V and 24 V · 6 and 10 N·m · 60 and 120 s



termojet.com.ua

WARNING! Read this manual before starting installation. Electrical connection must be carried out by a qualified electrician only.

Application

Termojet AQUA series electric actuators are intended for driving 3-way and 4-way rotary mixing valves with a 90° travel that require a torque of up to 6 N·m (up to 10 N·m for certain versions).

The actuator is fitted to Termojet RMV series valves through the adapter supplied with it. The slow travel — 60 or 120 seconds for 90° — rules out water hammer in the system.

Controls

- **Handle** with a position indicator on the top cover — shows the current angle of the valve.
- **Clutch button** beside the handle: press it and it latches down, releasing the handle for manual operation. Pressing it again returns the actuator to automatic mode.
- **DIP switch** inside the housing (modulating version only) — direction of rotation, signal type and running time.

Do not try to turn the handle without pressing the clutch button — this will damage the gearbox.

Technical data

Supply voltage	230 V AC 50–60 Hz or 24 V AC/DC
Power consumption	max. 5 W
Running time for 90°	60 or 120 s
Torque	6 N·m or 10 N·m
Angle of rotation	90°
Control signal	3-point, 2-point, 3-point + AUX, modulating 0(2)–10 V
Position indicator	yes, CW/CCW switchable
Manual operation	yes, by the clutch button
Cable	1 m, 3×0.75 mm ² or 4×0.5 mm ² , 6.5 mm diameter
Noise level	max. 40 dB at 100 mm
Protection rating	IP42
Conformity	CE, RoHS

Range

Part no.	Supply	Signal	Time	Torque
AQUAO401-3-230-120	230 V AC	3-point	120 s	10 N·m
AQUAO405-3-230-060	230 V AC	3-point	60 s	6 N·m
AQUAO412-3-230-060	230 V AC	2-point + AUX	60 s	6 N·m
AQUA411-0-10-24	24 V AC/DC	0(2)–10 V	60 / 120 s	10 N·m

A 10 N·m version of the 6 N·m models is available to order. The running time of the AQUA411 actuator is set by the DIP switch.

Dimensions

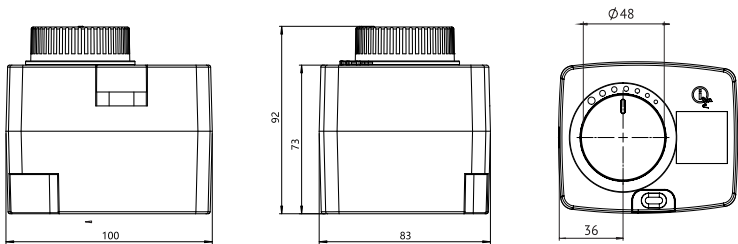


Fig. 1. Overall dimensions, mm

Mounting on the valve

1. Remove the handle of the valve and fit the actuator adapter onto the rotor spindle.
2. Fit the anti-rotation plate on the cover of the valve.
3. Turn the indicator of the adapter to the 45° position and make sure that the handle of the actuator is at 45° as well — see Fig. 2.
4. Secure the actuator, the handle and the position indicator with the M5×40 screw. The tightening torque must not exceed 2 N·m.

Mounting procedure

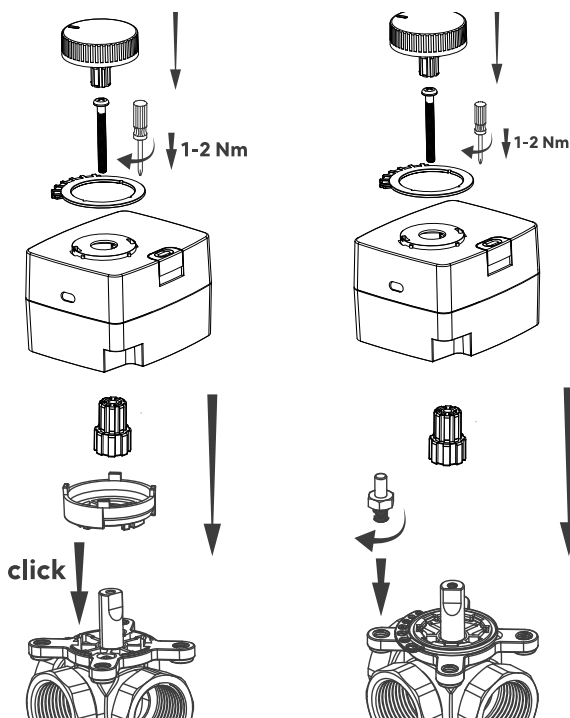


Fig. 2. Mounting the actuator on the valve step by step. A click of the catch means the adapter has seated in its working position

Operating the actuator with loose fixings is not permitted. The actuator may be mounted in any position except underneath the valve.

Disposal

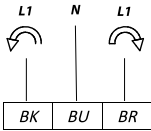
The product is marked with the crossed-out wheeled bin symbol. Under Directive 2002/96/EC and the legislation on waste electrical and electronic equipment, this marking means that at the end of its service life the product must not be disposed of together with household waste.

The user is obliged to hand the product over to a collection point for waste electrical and electronic equipment.

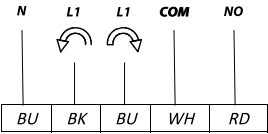
Wiring diagrams

Electrical connection must be carried out by a qualified electrician only, with the supply line isolated.

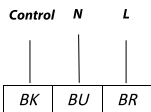
3-point control



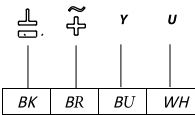
3-point + AUX



2-point control



Modulating 0(2)–10 V



Core colours: BK — black, BR — brown, BU — blue, WH — white, RD — red. The terminal markings are applied to the product itself and are not translated.

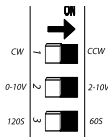


Fig. 3. DIP switch

DIP switch

Available on the modulating version. The factory positions are to the left:

- 1 — direction of rotation: CW (clockwise) or CCW;
- 2 — signal type: 0–10 V or 2–10 V;
- 3 — running time: 120 s or 60 s.

Product data

Product	Termojet AQUA series electric actuator
Manufacturer	Termojet
Distributor in Ukraine	Sofiivka-Montazh LLC, EDRPOU 37074476 EDRPOU — Ukrainian company registry code
Address	3 Kyivska St., Sofiivska Borshchahivka, Bucha district, Kyiv region, 08131, Ukraine
Contacts	+380 44 407 24 10 · termojet@sofiivka.kiev.ua termojet.com.ua

Storage and transport

Store the product in the manufacturer's packaging, in heated or unheated warehouses — no closer than one metre to any heating appliance — or under a canopy, in conditions that rule out mechanical damage.

Permissible storage and transport temperature is –20 to +50 °C. The product may be carried by any means of transport in accordance with the carriage rules in force for that mode of transport. During carriage, protect the product from impact, mechanical load and surface scratching.

Disposal

The product contains no substances hazardous to human life and health or to the environment. At the end of its service life, dispose of the product as household waste in accordance with local waste-management rules. Metal parts are suitable for remelting.

Acceptance and testing

The products covered by these instructions have been manufactured, tested and accepted in accordance with the manufacturer's current technical documentation.

Warranty

The manufacturer warrants that the product complies with safety requirements, provided that the user observes the rules set out in these instructions.

The service life of the product, provided that these instructions are observed and the required servicing is carried out, is **10 years** from the date the product is handed over to the user.

The warranty period is **24 months** from the date of sale, but no longer than the service life of the product.

The warranty covers all defects arising through the fault of the manufacturer and **does not cover** defects caused by:

- failure to observe the storage, installation, testing, operation or servicing conditions described here;
- improper transport or loading and unloading;
- exposure to substances aggressive to the materials of the product;
- fire, natural disaster or force majeure;
- acts of the user or third-party interference with the design of the product.

Products that fail within the warranty period because of a manufacturing defect are repaired or replaced free of charge. The cost of removing and transporting the faulty product is not reimbursed to the buyer. If a claim proves to be unfounded, the buyer pays for the diagnosis and examination.

To process a claim the following are required: a free-form statement giving contact details, the installation address and a description of the defect; proof of purchase; photographs of the faulty product, including photographs taken at the place of installation; a copy of the completed warranty card.

Claims are accepted at: 3 Kyivska St., Sofiivska Borshchahivka, Bucha district, Kyiv region, 08131, Ukraine. Phone +380 44 407 24 10, e-mail termojet@sofievka.kiev.ua.

The manufacturer reserves the right to introduce changes to the design of the «Termojet AQUA series electric actuator» product that do not impair its quality.

Warranty card

to delivery note No. _____ dated _____ 20____

Product: Termojet AQUA series electric actuator

No.	Part number	Quantity	Notes
1			
2			
3			
4			

Warranty period – 24 months from the date of sale.

I have read and accept the warranty terms and the installation and operating instructions:

Buyer (signature)

Seller (signature)

Date of sale

Stamp or seal
of the selling organisation